

Exercise 2. SQL Foundations

SQL Aggregate Functions & Operators

Question 1: The Students Table

- 1) List all distinct departments in the students table:

SQL Query:

```
SELECT DISTINCT department  
FROM students;
```

Output Table:

department
IT
HR
Finance

- 2) Get the average age of students per department.

SQL Query:

```
SELECT department, AVG(age) AS avg_age  
FROM students  
GROUP BY department;
```

Output Table:

department	avg_age
IT	21.5
HR	21.5
Finance	22.0

- 3) Show departments with more than 1 student.

SQL Query:

```
SELECT department, COUNT(student_id) AS student_count  
FROM students
```

GROUP BY department

```
HAVING COUNT(student_id) > 1;
```

Output Table:

department	student-count
IT	2
HR	2

4) Get all students whose age is between 21 and 23

SQL Query:

```
SELECT student_id, name, age, department  
FROM students
```

```
WHERE age BETWEEN 21 and 23;
```

Output Table:

student_id	name	age	Department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5) List all students in the IT or HR department who are older than 21

SQL Query:

```
SELECT student_id, name, age, department  
FROM students
```

```
WHERE department IN ('IT', 'HR')
```

```
AND age > 21;
```

Output Table:

student_id	Name	age	department
2	Bob	22	IT
4	Diana	23	HR

Question 3 : The courses Table

6) Show total credits per department only for departments with more than 5 total credits

SQL Query:

```
SELECT department, SUM(credits) AS total_credits  
FROM courses  
GROUP BY department  
HAVING SUM(credits) > 5;
```

Output Table:

department	total_credits
IT	11

7) List all courses that do not have 4 credits

SQL Query:

```
SELECT course_id, course_name, department, credits  
FROM courses  
WHERE credits <> 4;
```

Output Table:

Course_id	Course_name	department	Credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8) Show the top 3 courses by credits in descending order

SQL Query:

```
SELECT course_id, course_name  
FROM courses  
ORDER BY credits Desc  
Limit 3;
```


Output Table:

Course_id	Course_name
102	Python
103	Data Science
101	SQL Basics

Question 3 : The enrollments Table

9) Get the maximum, minimum and average grade across all enrollments.

SQL Query:

```
SELECT MAX(grade) AS max_grade, MIN(grade) AS min_grade, AVG(grade) AS avg_grade  
FROM enrollments;
```

Output Table:

max_grade	min_grade	avg_grade
90	78	84.6

10) Count how many enrollments exist per course

SQL Query:

```
SELECT course_id, COUNT(enrollment_id) AS enrollment_count  
FROM enrollments  
GROUP BY course_id;
```

Output:

Course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

Question 4 : The salaries Table

11) Find total salary and total bonus per department

SQL Query:

```
SELECT department, SUM(salary) AS total_salary,  
SUM(bonus) AS total_bonus  
FROM salaries  
GROUP BY department
```

Output Table:

department	total_salary	total_bonus
IT	122000	11000
HR	109000	7500
Finance	70000	6000

12) Show departments where average salary is above 55,000.

SQL Query:

```
SELECT department, AVG(salary) AS avg_salary  
FROM salaries  
GROUP BY department  
HAVING AVG(salary) > 55000
```

Output Table:

department	avg_salary
IT	62000
Finance	70000

13) List employees whose salary plus bonus is greater than 60,000.

```
SELECT employee_id, name, salary, bonus, (salary + bonus)  
AS total_compensation  
FROM salaries  
WHERE (salary + bonus) > 60000
```


Output Table:

employee_id	name	salary	bonus	total_compensation
1	Tom	60 000	5 000	65 000
3	Spike	70 000	6 000	76 000
4	Tyke	62 000	6 000	68 000

Question 5: The project Table

14) Show total and average budget per department. Only include departments with average budget above 70,000.

```
SELECT department, SUM(budget) AS total_budget,
AVG(budget) AS avg_budget
FROM Projects
GROUP BY department
HAVING AVG(budget) > 70 000;
```

Output Table:

department	total_budget	avg_budget
IT	270 000	135 000
Finance	80 000	80 000

15) List all projects with budgets between 50,000 and 120,000 excluding the Marketing department.

```
SELECT project_id, project_name, department, budget
FROM Projects
WHERE budget BETWEEN 50 000 AND 120 000
AND department <> 'Marketing';
```

Output Table:

project_id	project_name	department	budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
5	HR Portal	HR	50 000